When Accuracy Is Not Understanding: Auditing Annotation Shortcuts in a ClinVar-gnomAD Benchmark for AI-Assisted Variant Interpretation

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Abstract

Background. Computational predictions are used as supporting evidence when genetic variants are classified (ACMG/AMP criteria PP3 and BP4), and new genomic AI models are often validated by their ability to separate ClinVar pathogenic from gnomAD common variants. Whether such benchmarks measure understanding of the variant, or artefacts of how the data were assembled, is rarely tested.

Methods. We audited a published pathogenic-versus-common missense benchmark (22,254 pathogenic and 17,396 common variants in its annotation table; 3,000 per class sampled for modelling). Using identical leave-chromosome-group-out evaluation, we trained probes on (i) annotation metadata alone, with no sequence; (ii) frozen embeddings of the reference sequence alone; and (iii) the alternate-minus-reference embedding (the variant-specific signal) for DNABERT-2, HyenaDNA and a 4-mer baseline. AlphaGenome was evaluated separately on a 1,018-variant subset.

Results. Annotation metadata alone reached AUROC 0.968 (95% CI 0.963-0.972), higher than any sequence-based probe. The two classes use almost disjoint annotation vocabularies: none of the 22,254 pathogenic variants carries an upstream-, downstream- or regulatory-region term, and none of the common variants carries two terms found only in the pathogenic class. Restricted to terms used by both classes, annotation alone reached 0.624. Variant-specific embeddings were close to chance (AUROC 0.504-0.524 across models), whereas reference-only embeddings reached 0.568-0.681. Within plain-missense variants, where annotation carries little information, DNABERT-2 reference-only embeddings still reached 0.749 (95% CI 0.711-0.782), indicating a locus-level signal that annotation does not explain but that cannot distinguish alleles at the same position. AlphaGenome showed the same pattern.

Conclusions. On this benchmark, headline accuracy cannot be read as evidence that a model understands variant effects: benchmark-construction artefacts alone account for most of it. Because computational evidence enters clinical variant classification, we propose a seven-item reporting checklist for benchmarks of AI-assisted variant interpretation.

Keywords: variant interpretation; benchmark leakage; shortcut learning; ClinVar; gnomAD; ACMG/AMP; genomic foundation models; responsible AI

1 Introduction

Interpreting the clinical significance of genetic variants is a central bottleneck in genomic medicine. Most variants observed in patients are rare, and a large fraction remain variants of uncertain significance (Richards et al., 2015; Landrum et al., 2018). The ACMG/AMP guidelines allow computational predictions to contribute supporting evidence for pathogenicity (criterion PP3) or benignity (BP4), and the strength with which such evidence may be counted depends on how reliably a tool separates pathogenic from benign variation (Richards et al., 2015; Tavtigian et al., 2018; Pejaver et al., 2022). A tool whose measured accuracy is inflated therefore does not only mislead its developers: it can mis-weight evidence in the classification of real patients' variants.

Genomic foundation models, self-supervised neural networks trained on DNA sequence, are now proposed as general-purpose engines for this task, and are routinely compared by how well a simple classifier on their embeddings separates ClinVar pathogenic variants from common variants in gnomAD (Feng et al., 2025; Benegas et al., 2023; Zhou et al., 2024; Nguyen et al., 2023; Avsec et al., 2026). Such a benchmark is only informative if the two classes differ *only* in the property of interest. In practice, pathogenic and common variants are drawn from different databases, curated by different processes, in different genes, and annotated by pipelines that may differ. Evaluations of missense-effect predictors have long been recognised to be vulnerable to circularity of this kind, in which a model can score well by recognising where a variant sits rather than what it does (Grimm et al., 2015); the more general phenomena are known as shortcut learning and data leakage (Geirhos et al., 2020; Kapoor and Narayanan, 2023).

We therefore ask a question about the benchmark rather than about any single model: *can benchmark construction and annotation artefacts create the appearance that genomic AI understands variant pathogenicity?* We make three contributions. (i) We show that annotation metadata alone, with no sequence, separates the classes of a published pathogenic-versus-common missense benchmark far better than any sequence embedding we tested, and we trace this to near-disjoint annotation vocabularies. (ii) We use three models (DNABERT-2, HyenaDNA and a 4-mer baseline; AlphaGenome on a separate subset) as supporting evidence, using a reference-only control that cannot see the variant and a stratified analysis that removes most annotation differences. (iii) We translate the findings into a seven-item reporting checklist and discuss what they imply for the use of computational evidence in variant classification. We deliberately make no claim about which model is best, and we do not claim to establish a clinically usable classifier.

2 Methods

2.1 Study design

Figure 1 summarises the design. All analyses use frozen embeddings or tabular metadata; no model was fine-tuned and no new embeddings were generated for this study; all embeddings had

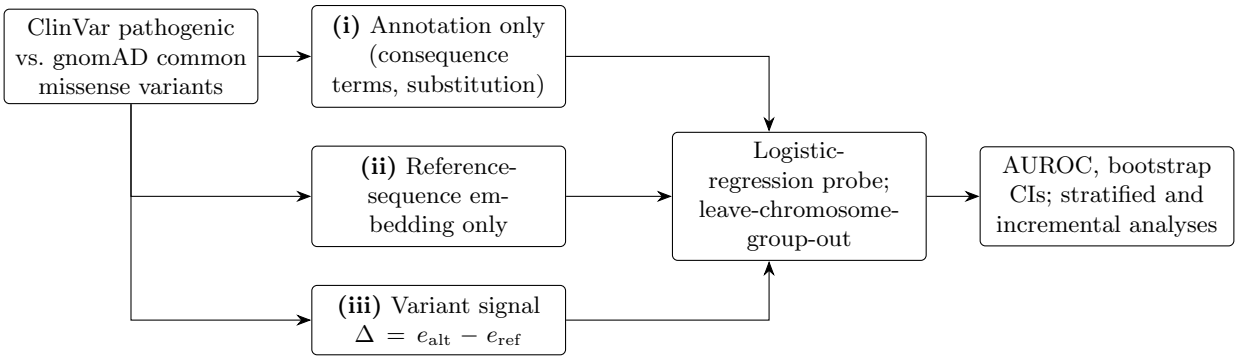


Figure 1: Study design. The same variants, folds and bootstrap resamples are used for the three feature sources. Source (i) contains no sequence; source (ii) cannot see the alternate allele; only source (iii) isolates the effect of the substitution.

been computed earlier.

2.2 Benchmark and annotation data

Variants come from the pathogenic-variant benchmark of Feng et al. (2025), derived from the Genomics Long-Range Benchmark (Kao et al., 2024): ClinVar pathogenic variants (Landrum et al., 2018) versus gnomAD common variants (Karczewski et al., 2020) (minor allele frequency 0.05-0.50), single-nucleotide variants on GRCh38 with a missense consequence in at least one transcript. The accompanying annotation table lists, for each variant, a consequence string, ClinVar review status and genomic coordinates (22,254 pathogenic and 17,396 unique common variants after removing duplicate coordinates). We drew a class-balanced random sample of 3,000 variants per class (seed 42; autosomes and X; duplicates removed) and merged annotation fields on chromosome, position, reference and alternate allele; all 6,000 sampled variants were matched. The sampling is balanced by design, so accuracy and F1 are not reported; AUROC is used throughout.

2.3 Annotation-only probes

To measure how much of the task can be solved without sequence, we trained logistic-regression probes on annotation metadata only: (a) a single indicator for "plain missense" (consequence string exactly `missense_variant`); (b) two harmonised indicators (plain missense; any non-coding-transcript term, ignoring spelling); (c) one indicator per consequence term as written; (d) (c) plus the nucleotide substitution type; and (e) consequence terms observed in at least 1% of *both* classes in the full annotation table (shared vocabulary). Minor allele frequency and allele counts were never used as features, because they are populated only for the common class and so identify the label by construction.

2.4 Sequence-embedding probes

Each model embedded the identical 501-bp window centred on the variant (reference and alternate alleles); last-layer hidden states were mean-pooled. Models were DNABERT-2 (Zhou et al., 2024), HyenaDNA (Nguyen et al., 2023) and a 4-mer frequency baseline requiring no pretraining. These embeddings had already been computed for an earlier benchmark; no HyenaDNA or DNABERT-2 computation was repeated. Two feature sources were evaluated for each model: the reference embedding alone (a context-only control, Hewitt and Liang 2019), and the difference $\Delta = e_{\text{alt}} - e_{\text{ref}}$, which isolates the effect of the substitution on the representation. AlphaGenome (Avsec et al., 2026), run through the `alphagenome-pytorch` port (Bredikhin et al., 2026) (version 0.3.1; all-folds weights, file `model_all_folds.safetensors`), had been embedded only for the 1,018 variants of the upstream test split (461 pathogenic, 557 common), with a 196,608-bp N-padded input and mean pooling; it is analysed separately and never pooled with the other models.

2.5 Probe and evaluation

Features were standardised with training-fold statistics and classified with L2-regularised logistic regression ($C = 1$; scikit-learn, Pedregosa et al. 2011), identically for every feature source. Variants were held out by chromosome group (three groups as in Feng et al. 2025), so linked neighbouring variants never straddle training and test; out-of-fold predictions were pooled. For the AlphaGenome subset, which lies entirely on chromosome 8, five contiguous position blocks were used instead; this limits but does not remove linkage between folds. Uncertainty was quantified by class-stratified bootstrap (1,000 resamples; Efron and Tibshirani 1993) and, as a sensitivity analysis, by resampling whole chromosomes. A label-shuffling control (20 permutations) checked the pipeline against chance.

2.6 Stratified and incremental analyses

To remove most of the annotation difference, we repeated the evaluation within strata defined by consequence annotation: plain missense, non-coding-transcript flagged, and other. Scores were evaluated either from the probe trained on all variants or from a probe trained and tested within the stratum. To test whether sequence adds information beyond annotation, we compared an annotation-only model with the same model plus the out-of-fold reference-only score (paired bootstrap of the AUROC difference). Sensitivity to ClinVar label quality used the review status (assertion criteria provided; multiple submitters or expert panel), keeping all common variants.

2.7 Reproducibility and ethics

All steps are scripted and seeded (seed 42); code, per-variant out-of-fold scores and every result table are available at `https://github.com/willhhx/variant-benchmark-audit`. The study uses public, de-identified data and involved no human participants.

Consequence term (as annotated)	Pathogenic (%)	Common (%)
missense_variant	100.0	95.2
intron_variant	7.6	30.6
5_prime_UTR_variant	4.2	9.0
3_prime_UTR_variant	1.2	5.8
non-coding_transcript_variant	13.5	0.0
initiatior_codon_variant	3.0	0.0
non_coding_transcript_exon_variant	0.0	47.9
downstream_gene_variant	0.0	44.7
upstream_gene_variant	0.0	40.0
regulatory_region_variant	0.0	30.2

Table 1: Consequence terms by class in the full annotation table (percentage of variants in each class carrying the term; a variant can carry several terms). Terms are reproduced exactly as annotated, including one spelling error in the source.

3 Results

3.1 The two classes are annotated with almost disjoint vocabularies

In the annotation table, 76.0% of pathogenic variants (16,903) but only 8.5% of common variants (1,479) have the consequence string `missense_variant` with no other term (Table 1). Several terms occur in one class only: none of the pathogenic variants carries `non_coding_transcript_exon_variant`, `downstream_gene_variant`, `upstream_gene_variant` or `regulatory_region_variant` (versus 8,339, 7,779, 6,959 and 5,247 common variants), while the terms `non-coding_transcript_variant` (2,996 pathogenic variants) and `initiatior_codon_variant` (663; spelled as in the source) never occur in the common class. The class-specific spellings are consistent with the two classes having been annotated by different tools or versions, which we could not confirm from the files available.

3.2 Annotation alone outperforms every sequence embedding

A probe trained on annotation metadata and substitution type, with no sequence, reached AUROC 0.968 (95% CI 0.963-0.972) (Table 2; Figure 2A). Even a single indicator for plain missense reached 0.812 (95% CI 0.801-0.823), and two harmonised flags reached 0.835. In contrast, the best sequence-based probe (DNABERT-2 reference-only) reached 0.681 (95% CI 0.668-0.694). Most of the annotation advantage came from class-specific vocabulary: restricted to the four terms used by both classes, annotation alone reached 0.624 (95% CI 0.609-0.637), comparable to HyenaDNA reference-only (0.617). Within the non-coding-transcript stratum (424 pathogenic, 1,897 common) the annotation strings alone separated the classes perfectly (AUROC 1.000), because the pathogenic class uses a different spelling of the non-coding term.

Feature source	AUROC (95% CI)	Chromosome-cluster 95% CI
Annotation: consequence terms + substitution	0.968 (0.963-0.972)	0.961-0.974
Annotation: harmonised flags (2)	0.835 (0.824-0.845)	0.812-0.862
Annotation: plain-missense flag only	0.812 (0.801-0.823)	0.789-0.842
DNABERT-2, reference-only	0.681 (0.668-0.694)	0.660-0.703
DNABERT-2, variant (Δ)	0.507 (0.492-0.522)	0.491-0.524
HyenaDNA, reference-only	0.617 (0.604-0.632)	0.602-0.636
HyenaDNA, variant (Δ)	0.524 (0.508-0.537)	0.501-0.546
4-mer, reference-only	0.568 (0.554-0.583)	0.551-0.585
4-mer, variant (Δ)	0.504 (0.490-0.519)	0.479-0.527
Substitution type only (no annotation)	0.580 (0.565-0.594)	0.552-0.603

Table 2: Main results (6,000 variants; leave-chromosome-group-out; pooled out-of-fold AUROC). Annotation rows use no sequence. Reference-only embeddings cannot see the alternate allele. The variant (Δ) rows isolate the effect of the substitution.

3.3 Variant-specific embeddings are close to chance

The variant-specific (Δ) probes reached AUROC 0.507 (DNABERT-2), 0.524 (HyenaDNA) and 0.504 (4-mer), all within about 0.03 of chance, and did not rise above chance in any stratum (Table 3). Label-shuffling gave a mean AUROC of 0.501 (maximum 0.519 over 40 permutations), so the pipeline itself is not producing spurious signal. The median relative size of the alternate-minus-reference difference was 0.525% (HyenaDNA), 4.05% (DNABERT-2) and 0.00867% (AlphaGenome) of the reference embedding norm, so a one-base change is a small perturbation of a pooled representation, a point we return to below.

3.4 What remains once annotation is removed

Within plain-missense variants, annotation metadata carries little information (substitution type alone gives AUROC 0.582 there), yet reference-only embeddings remained informative: DNABERT-2 0.749 (95% CI 0.711-0.782), HyenaDNA 0.635 (95% CI 0.596-0.674) and the 4-mer baseline 0.552 (Table 3; Figure 2B). Probes trained within the stratum gave similar values (0.721 and 0.649). Adding the reference-only score to annotation raised AUROC only slightly (DNABERT-2 by 0.010 (95% CI 0.007-0.013); HyenaDNA by 0.005 (95% CI 0.002-0.007)). A signal tied to the sequence *around* the variant therefore persists, but it is not the variant: by construction a reference-only score is identical for different alleles at the same position.

3.5 Robustness

Chromosome-cluster intervals were similar to the bootstrap intervals (Table 2) and fold-level AUROCs were consistent across the three chromosome groups (Supplementary Table S1). Restricting pathogenic variants to those with assertion criteria, or to multiple-submitter and expert-panel assertions, left every result essentially unchanged (annotation alone 0.965-0.968; Supplementary

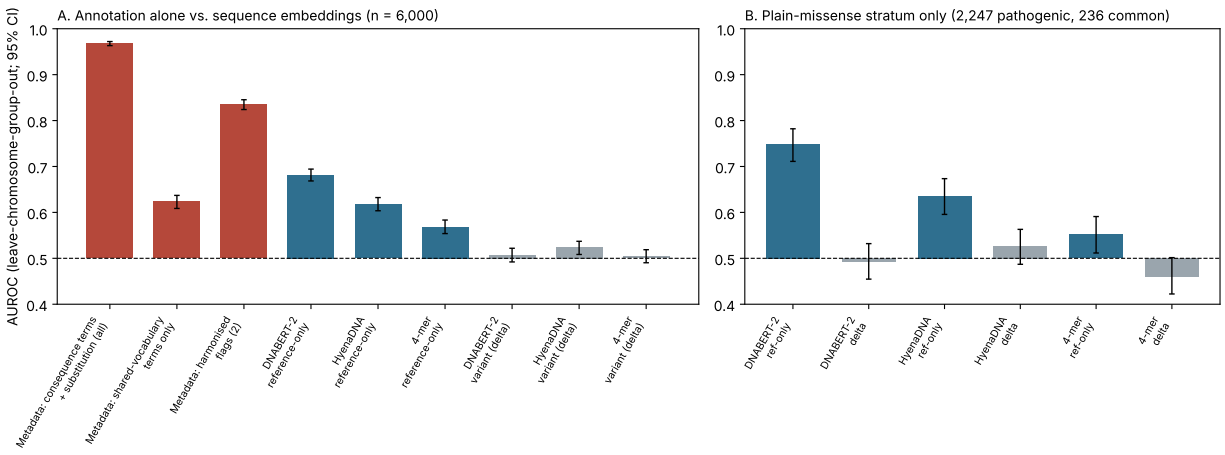


Figure 2: (A) AUROC of annotation-only probes (red), reference-only embeddings (blue) and variant (Δ) embeddings (grey) for all 6,000 variants; error bars are 95% bootstrap intervals. (B) The same embedding probes restricted to plain-missense variants (2,247 pathogenic, 236 common), where annotation carries little information. The dashed line marks chance.

Table S3), so the findings are not driven by the weakest ClinVar assertions. The annotation result (AUROC 0.964-0.968) and the near-chance variant results (0.496-0.527) were stable across probe regularisation strengths, whereas reference-only AUROC varied (DNABERT-2 0.680-0.712; HyenaDNA 0.556-0.617; Supplementary Table S2); we report $C = 1$ throughout, chosen in advance, and none of the conclusions depends on this choice.

3.6 AlphaGenome subset

In the separate 1,018-variant subset, annotation alone again reached 0.959 (95% CI 0.945-0.970), the reference-only embedding 0.608 (95% CI 0.573-0.643) and the variant (Δ) embedding 0.514 (95% CI 0.481-0.552) (Table 4). These numbers are not comparable with the other models, because the subset is small, lies entirely on one chromosome and uses position-block folds; we report it as an independent confirmation of the pattern, not as a ranking.

4 Discussion

Principal findings. In a widely used style of benchmark, annotation metadata with no sequence separates ClinVar pathogenic from gnomAD common missense variants almost perfectly (AUROC 0.968), whereas the effect of the substitution on model embeddings is close to chance. The headline accuracy of a sequence-based classifier on this task can therefore not be taken as evidence that it understands variant effects. A residual locus-level signal exists in reference-sequence embeddings, most clearly for DNABERT-2, but it cannot by itself discriminate between variants at the same locus.

Feature source	Scores from probe trained on	AUROC (95% CI)
Annotation (substitution type only here)	all 6,000 variants	0.582 (0.545-0.620)
DNABERT-2 reference-only	all 6,000 variants	0.749 (0.711-0.782)
DNABERT-2 reference-only	plain-missense variants only	0.721 (0.683-0.757)
DNABERT-2 variant (Δ)	all 6,000 variants	0.493 (0.455-0.532)
HyenaDNA reference-only	all 6,000 variants	0.635 (0.596-0.674)
HyenaDNA reference-only	plain-missense variants only	0.649 (0.609-0.688)
HyenaDNA variant (Δ)	all 6,000 variants	0.527 (0.487-0.563)
4-mer reference-only	all 6,000 variants	0.552 (0.512-0.591)
4-mer variant (Δ)	all 6,000 variants	0.460 (0.422-0.501)

Table 3: Plain-missense stratum (2,247 pathogenic, 236 common). With only 236 common variants, intervals are wide and the stratum should be read as supporting, not definitive, evidence. The annotation row reflects substitution type only because consequence strings are identical within the stratum. All strata are in Supplementary Table S4.

Feature source (1,018 variants; position-block CV)	AUROC (95% CI)
Annotation: consequence terms + substitution	0.959 (0.945-0.970)
AlphaGenome reference-only	0.608 (0.573-0.643)
AlphaGenome variant (Δ)	0.514 (0.481-0.552)

Table 4: AlphaGenome validation subset (461 pathogenic, 557 common variants; all on chromosome 8; five position-block folds). Not comparable with Table 2.

Why shortcut-driven accuracy is misleading for computational evidence. The PP3 and BP4 criteria assume that a tool's output reflects something about the variant's effect, and the strength of evidence assigned to a score depends on its calibration against variants not used to build the tool (Richards et al., 2015; Pejaver et al., 2022). Shortcut-driven accuracy undermines this in three ways. First, a shortcut that is present in a benchmark is typically absent in a diagnostic laboratory, where every candidate variant has been annotated by the same pipeline; the performance measured on the benchmark then overstates the performance obtainable in the clinic. Second, evidence strength is derived from measured accuracy, so inflated accuracy leads to over-weighted evidence. Third, features that correlate with annotation (for example, transcript type or gene) are partly information the classifier already has, creating a risk of counting the same evidence twice. The reference-only result is the genomic analogue of a well-described problem: tools can learn which genes or loci tend to harbour pathogenic variants rather than which variants are pathogenic (Grimm et al., 2015). The clinically hard problem, deciding which variant *within* a gene is pathogenic, is exactly the one a reference-only score cannot address.

Reporting checklist. The most useful takeaway of this audit is practical. Table 5 lists seven items that a benchmark of AI-assisted variant interpretation should report. They are inexpensive, require no new model, and align with general reporting guidance for prediction models that use machine learning (Collins et al., 2024).

Implications for public health. Variant classification underpins diagnosis, cascade testing of

	Report	Why (finding in this study)
1	An annotation-only baseline built from every metadata field available for both classes	Annotation alone reached AUROC 0.968 versus 0.681 for the best embedding
2	A class-by-term table of annotation vocabulary, and confirmation that both classes were annotated by one pipeline and version	Several terms occurred in one class only, including a misspelt term
3	A context-only (reference-sequence) control	Reference-only embeddings out-performed variant-specific ones by up to 0.174 AUROC
4	The variant-specific signal on its own (Δ or an equivalent), reported next to the headline metric	Variant-specific embeddings were within about 0.03 of chance
5	Results in matched or stratified subsets, with per-class counts and intervals	Plain-missense stratum has only 236 common variants; conclusions there are tentative
6	Held-out chromosomes or loci, with intervals that account for clustering	Chromosome-cluster intervals were reported alongside bootstrap intervals
7	The intended use, the population in which accuracy was measured, and an explicit statement that benchmark accuracy is not evidence for clinical use	The benchmark is balanced and retrospective

Table 5: Reporting checklist for benchmarks of AI-assisted variant interpretation.

relatives and reproductive decisions, and reclassification is costly for patients and laboratories. As AI-assisted tools enter these workflows, reliability depends on evaluations that are transparent and reproducible, and on users being able to see what a reported accuracy does and does not demonstrate. Our results argue for independent, benchmark-agnostic validation before computational scores are given weight as evidence, and for benchmark maintainers to annotate all classes with a single pipeline. Equitable access to reliable genetic testing depends on evaluations that hold across patient populations. By showing how a single annotation pipeline difference can inflate apparent accuracy, this audit gives developers and clinical laboratories a concrete check to run before a tool is deployed for all patients, and the reporting checklist asks authors to state the population in which accuracy was measured, so that readers can judge whether results apply to the patients in front of them.

Relation to prior work. Large benchmarking efforts report that model rankings are unstable and that representations depend on the downstream task and on how embeddings are summarised (Feng et al., 2025; Tang and Koo, 2025; Ledneva et al., 2026). Our work is complementary: it does not rank models, and instead asks whether the benchmark measures what it is taken to measure. The control we use follows the logic of control tasks for probes (Hewitt and Liang, 2019).

Limitations

(1) We examined one benchmark and one task; similar audits are needed for other variant benchmarks. (2) We used a class-balanced sample of 3,000 variants per class, which does not reflect clinical prevalence. (3) The plain-missense stratum contains only 236 common variants (48 in the AlphaGenome subset), and those variants are not a random sample of common variation, so stratified estimates are imprecise. (4) We could not establish why the vocabularies differ; the pattern is consistent with different annotation tools or versions, which should be checked with the benchmark's maintainers. (5) We could not identify the source of the residual reference-only signal. Candidate explanations include gene identity, evolutionary constraint, sequence composition and other dataset differences; the annotation table has no gene identifiers, so a gene-held-out analysis was not possible. (6) Frozen, mean-pooled embeddings and linear probes measure information that is easily accessible, not what a model encodes. The variant perturbation is small (0.525%, 4.05% and 0.00867% of the embedding norm for HyenaDNA, DNABERT-2 and AlphaGenome), so near-chance variant signal may reflect pooling dilution rather than an absence of variant information; pooling at the variant position and using intermediate layers were not tested, and we therefore do not claim that these models cannot represent variants. (7) The AlphaGenome subset is small, lies on one chromosome and uses position-block folds. (8) Three models were examined, chosen because their embeddings already existed; the conclusions concern the benchmark, not model architectures. (9) Nothing here has been validated clinically or in diverse populations, and none of the models or probes is intended for clinical use.

5 Conclusion

On a published pathogenic-versus-common missense benchmark, annotation metadata alone, with no sequence, performed far better than genomic AI embeddings, and variant-specific embeddings performed near chance. Benchmark accuracy of this kind is therefore not evidence of understanding variant effects, and it should not be used to justify weight for computational evidence in variant classification. We recommend that benchmarks of AI-assisted variant interpretation report an annotation-only baseline, a context-only control and stratified results, as set out in the checklist above.

Data and code availability

Variants derive from the public benchmark of Feng et al. (2025). Code, per-variant out-of-fold scores and all result tables are available at `https://github.com/willhhx/variant-benchmark-audit`.

References

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, Kyle R. Taylor, Tom Ward, Clare Bycroft, Lauren Nicolaisen, Eirini Arvaniti, Joshua Pan, et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649(8099):1206–1218, 2026.

Gonzalo Benegas, Sanjit Singh Batra, and Yun S. Song. DNA language models are powerful predictors of genome-wide variant effects. *Proceedings of the National Academy of Sciences*, 120 (44):e2311219120, 2023.

Danila Bredikhin, Martin Kjellberg, Christopher Zou, Alejandro Buendia, Xinming Tu, and Anshul Kundaje. alphagenome-pytorch: a PyTorch port of AlphaGenome. https://github.com/genomicsxai/alphagenome-pytorch, 2026. Python package `alphagenome-pytorch`, Apache-2.0; model parameters and outputs remain subject to the Google DeepMind AlphaGenome Model Terms.

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

Haonan Feng, Lang Wu, Bingxin Zhao, Chad Huff, Jianjun Zhang, Jia Wu, Lifeng Lin, Peng Wei, and Chong Wu. Benchmarking DNA foundation models for genomic and genetic tasks. *Nature Communications*, 16:10780, 2025. doi: 10.1038/s41467-025-65823-8.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

Dominik G. Grimm, Chloé-Agathe Azencott, Fabian Aicheler, Udo Gieraths, Daniel G. MacArthur, Kaitlin E. Samocha, David N. Cooper, Peter D. Stenson, Mark J. Daly, Jordan W. Smoller, Laramie E. Duncan, and Karsten M. Borgwardt. The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation*, 36(5): 513–523, 2015.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, 2019.

Chia-Hsiang Kao et al. Advancing DNA language models: The genomics long-range benchmark. In *ICLR 2024 Workshop on Machine Learning for Genomics Explorations*, 2024.

Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

Konrad J. Karczewski et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581:434–443, 2020.

Melissa J. Landrum et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018.

Daria Ledneva, Mikhail Nuridinov, and Denis Kuznetsov. GENEB: Why genomic models are hard to compare, 2026. arXiv:2606.04525.

Eric Nguyen et al. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Vikas Pejaver, Alicia B. Byrne, Bing-Jian Feng, Kymberleigh A. Pagel, Sean D. Mooney, Rachel Karchin, Anne O'Donnell-Luria, Steven M. Harrison, Sean V. Tavtigian, Marc S. Greenblatt, Leslie G. Biesecker, Predrag Radivojac, and Steven E. Brenner. Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *American Journal of Human Genetics*, 109(12):2163–2177, 2022.

Sue Richards et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*, 17:405–424, 2015.

Ziqi Tang and Peter K. Koo. Evaluating the representational power of pre-trained DNA language models for regulatory genomics. *Genome Biology*, 2025. doi: 10.1186/s13059-025-03674-8.

Sean V. Tavtigian, Marc S. Greenblatt, Steven M. Harrison, Robert L. Nussbaum, Snehit A. Prabhu, Kenneth M. Boucher, and Leslie G. Biesecker. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genetics in Medicine*, 20(9): 1054–1060, 2018.

Zhihan Zhou et al. DNABERT-2: Efficient foundation model and benchmark for multi-species genomes. In *International Conference on Learning Representations (ICLR)*, 2024.